Phase2v1: This phase was simply to set up the code for later structuring. Looking back (just now) – it seems that I overwrote v2 with v1, so v1 no longer truly exists. Luckily, the v1 version was fairly miserable and filled with simple syntax problems which are a bit embarrassing – but this is still not very helpful for reporting documentation. Oops! I will attempt to paint the picture of what existed in v1 below.

imageUpscale(), loadPhoto(), and main existed, with their contents mostly bare. Include statements existed but one was missing. Main included some incorrectly written function calls. – That was about it. I really only spent about 10 minutes on this version.

Time Completed: March 11, 2026 – around 8pm.

Phase2v2: Most of the work was done here. Lab 3 code was taken directly from the lab documentation, as to avoid figuring out how to deal with BMP headers, creation, and loading. imageUpscale and loadPhoto were changed to accepted a created image in the heap, rather than the older version's int[] (terrible) attempt. Created the upscaler as an image pixel doubler, meaning every pixel is multiplied four times. Loading the photo took some time and research, getting me to blend what I could find online with the Lab3 code, which seemed to work well. Added atoi statements in main, free'd data, fixed function calls.

This version took me about an hour and a half, maybe two hours.

Time Completed: March 12, 2025 – around 7pm.

Phase2v3: Debugging! Luckily, not many problems arose from the constructed code. I got confused from the padSize for a bit and played around with that. I also got confused about the BGR -> RGB file formatting, and had to play around with that (incorrectly, lab3 code was fine.) Altogether, this project submission was not very big, so it was fairly easy to complete without a ton of version histories (coding time) required.

Time Completed: March 12, 2026 – around 8:30pm. Some time between v2 and v3 was to create the AI generated photo sample and print it as a bmp file.

Compilation Screenshot:

```
PS C:\Users\Jordan\Desktop\Winter 2026 Courses\Parallel Processing\Project\Phase2code> gcc Phase2v3.c -o upscaler
PS C:\Users\Jordan\Desktop\Winter 2026 Courses\Parallel Processing\Project\Phase2code> ./upscaler 1024 1024
File open failed: No such file or directory
PS C:\Users\Jordan\Desktop\Winter 2026 Courses\Parallel Processing\Project\Phase2code> ./upscaler 1024 1024
Saved projectHelloWorld.bmp
```

Input Image:



Created Image:

